CS/SE 4X03 SCIENTIFIC COMPUTATION

NED NEDIALKOV
DEPARTMENT OF COMPUTING AND SOFTWARE
MCMASTER UNIVERSITY
FALL 2023

Lectures:	Monday, Thursday	9:30AM–10:20AM	JHE	376
	Tuesday	10:30AM–11:20AM	JHE	376
Tutorials:	Thursday	10:30AM–11:20AM	JHE	326H
	Thursday	11:30AM–12:20PM	JHE	326H
	Thursday	1:30PM–2:20PM	JHE	326H
	Thursday	2:30PM–3:20PM	JHE	326H
Office hours:	Tuesday	11:30-12:30, 2:30-3:30	ITB	123

Instructor

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1. Introduction

This is an introductory course to scientific computing. We will study floating-point issues in numerical computing and various numerical algorithms. We will learn about numerical differentiation and integration, solving linear and nonlinear systems, computing eigenvalues, least-squares problems, and numerical methods for ordinary differential equations. This course also gives a brief introduction to deep learning. We will study error analysis, convergence, and stability of numerical algorithms. We will use Matlab as a programming language and environment.

2. Learning objectives

Postcondition A *learning objective* for a course is something the student is expected to know and understand or to be able to do by the end of the course. The learning objectives for this course are given below. Taken together, this set of learning objectives constitute the *postcondition* of the course.

1. Students should know and understand
   (a) issues in floating-point (FP) computations, overflow, cancellations, roundoff errors
   (b) truncation errors
   (c) solving linear systems
   (d) linear least squares

(e) solving non-linear equations
(f) interpolation
(g) numerical integration
(h) stability of methods for initial-value problems in ordinary differential equations

2. Students should be able to
   (a) perform roundoff error analysis
   (b) derive error bounds
   (c) interpret numerical results
   (d) perform simple complexity analysis
   (e) analyze convergence
   (f) analyze stability
   (g) write Matlab programs implementing numerical methods

Precondition The *precondition* of the course is the set of university-level learning objectives that the student is expected to have achieved before the start of the course.

The precondition includes knowledge of calculus and linear algebra

3. Materials

- Required: Ward Cheney and David Kinkaid. Numerical Mathematics and Computing
- Optional: Student Solution Manual
- McMaster standard calculator will be needed for the midterm and the final.

4. Topics

- Floating-point arithmetic
- Interpolation and numerical differentiation
- Numerical integration
- Solving systems of linear equations
- Computing eigenvalues
- Least squares
- Solving nonlinear equations
- Introduction to machine learning
- Methods for initial value problems for ordinary differential equations

5. Reading and suggested exercises

These may be updated as we progress with the course.
§X: Y means Section X should be studied and exercise Y is suggested
The corresponding numbers from the 6th edition are given in $(\cdots)$.

- Chapter 1
  - §1.1: 8, 10 (§1.1: 8, 10)
  - §1.2: 8, 9, 23, 30, 41 (§1.2: 8, 9, 23, 30, 41)
  - §1.3: 13, 14, 18, 22, 24, 45 (§2.1: 13, 14, 18, 24, 45)
  - §1.4: 8, 11, 13, 14, 24, 26, 29 (§2.2: 8, 11, 14, 24, 26, 29)
    Note: you can skip the theorem on loss of precision
- Chapter 2
  - §2.1: 1, 2, 7a (§7.1: 1, 2, 7a)

  - §2.2: 1, 8, 9, 23 (§7.2: 1, 8, 9, 23)
- Chapter 8
  - §8.1, pp. 358–365, 371–373: 2, 4, 11, 15, 18, 23 (§8.1, pp. 293–302, 306–307: 2, 4, 11, 15, 18, 23)
- Chapter 4
  - §4.1, pp. 153–168: 1, 9, 12, 18, 40 (§4.1, pp. 124–141: 1, 9, 12, 18, 40)
  - §4.2: 5, 9, 10, 15 (§4.2: 5, 9, 10, 15)
- Chapter 5
  - §5.1, up to p. 210: 1, 6, 7, 8, 11 (§5.2, up to p. 196: 2, 4, 5, 7, 19)
  - §5.3: 1, 2, 4 (§6.1: 1, 2, 4)
- Chapter 9
  - §9.1: 3, 5, 13, 25 (§12.1: 3, 5, 13, 25)
- Chapter 3
  - §3.2: 1, 3, 13, 14, 23 (§3.2: 1, 3, 13, 14, 23)
  - §3.1: 8, 12, 13 (§3.1: 8, 12, 13)
  - §3.3: 2, 3 (§3.3: 2, 3)
- Chapter 8
- Chapter 7

6. Grading scheme

These are tentative dates and may change depending on how we progress with the material.

Assignment 1	8%	21 September – 5 October
Assignment 2	8%	5 October – 25 October
Midterm	20%	30 October, during class time
Assignment 3	8%	2 November – 16 November
Assignment 4	8%	16 November – 30 November
Final exam	48%	

7. Course Policy

Lectures

- All course material will be posted on Avenue.

Assignments

- Late work will not be graded without an MSAF.

Missed work

- The MSAF accommodation for a missed assignment is a **three calendar day** extension from the original assignment deadline.
- The MSAF accommodation for a missed midterm is to roll the weight of the midterm into the weight of the final examination.

Remarking

- Requests for remarking of an assignment or a test must be made within one week after the marked assignment/test is returned.
- Requests that are later than a week will not be accommodated.

Changes

- The instructor reserves the right to modify elements of this course and will notify students accordingly.

8. Resources

- Get Started with MATLAB
- Floating Points
- D. Goldberg. What Every Computer Scientist Should Know About Floating-Point Arithmetic

9. Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy. The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained
- improper collaboration in group work
- copying or using unauthorized aids in tests and examinations

10. Authenticity/Plagiarism detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty. Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of please go to Turnitin.com.

11. Faculty Notices

"The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer or the Human Rights Consultant, as the problem occurs."